

On the Ad Hoc Status of Bartlett Corrections for the Satorra–Bentler Statistic

A Theoretical Investigation with Proof of Rejection

Based on: Savalei (2010), Satorra & Bentler (1994), Yuan & Bentler (1998),
Yuan, Yang & Jiang (2017), Wakaki, Eguchi & Fujikoshi (1990),
and the Cordeiro–Cribari-Neto General Framework

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Contents

1	Introduction: The Problem	3
2	Background: Two Statistics with Different Origins	3
2.1	The ML Statistic T_{ML} and Bartlett Correctability	3
2.2	The Satorra–Bentler Statistic T_{SC}	4
2.3	Savalei’s Ad Hoc Proposal (2010)	4
3	The Theoretical Gap: Why the Transfer Fails	4
3.1	Stage 1: Bias of T_{ML} Under Normality vs. Nonnormality	4
3.2	Stage 2: The Bias Decomposition of T_{SC}	5
3.3	Stage 3: Structural Incompatibility	6
4	Formal Proof: The Ad Hoc Transfer Is Invalid	6
4.1	Main Result	6
4.2	Corollary: T_{SC} Is Not Bartlett-Correctable	7
4.3	Corollary: No Sufficient Conditions Exist Short of Normality	7
5	When Does the Ad Hoc Correction Work by Accident?	7
6	Literature Evidence Consistent with the Rejection	8
7	What Does Have Theoretical Justification?	8
8	Conclusion: The Verdict	9

1 Introduction: The Problem

In structural equation modeling (SEM), the Satorra–Bentler scaled chi-square statistic T_{SC} is the standard test statistic for assessing model fit when data deviate from multivariate normality. Although T_{SC} is asymptotically χ^2_{df} -distributed under mild regularity conditions, its finite-sample performance can be poor: the Type I error rate may deviate substantially from the nominal level α , particularly when the sample size n is small relative to the number of observed variables p .

Savalei (2010) proposed applying the Bartlett (1950) correction factor—originally derived for the ML statistic T_{ML} under normality—to T_{SC} . Specifically, her Equations 7 and 8 multiply T_{SC} by correction factors derived under normal-theory assumptions. Savalei herself acknowledges that these corrections are “ad hoc,” stating that the correct Bartlett adjustment for general covariance structures was derived by Wakaki, Eguchi, and Fujikoshi (1990) and is “extremely complicated.”

The central question of this monograph is: **Can the ad hoc label be removed? Is there a theoretical basis—or a set of sufficient conditions—under which the normal-theory Bartlett correction validly transfers to T_{SC} ?**

We answer this question definitively in the negative. Through a careful analysis of the Edgeworth expansions of T_{ML} and T_{SC} , we prove that the n^{-1} bias structures of the two statistics are fundamentally different under nonnormality, and that no conditions short of normality itself can make the transfer valid. We further show that the problem is not merely one of “complicated formulas” but reflects a deep structural incompatibility between the Bartlett correction mechanism and the ratio structure of T_{SC} .

Our findings have practical implications: practitioners who use Savalei’s Equations 7 or 8 should understand that these corrections lack theoretical justification and may produce misleading results. We identify the alternative approaches that *do* have theoretical grounding and recommend their adoption.

2 Background: Two Statistics with Different Origins

2.1 The ML Statistic T_{ML} and Bartlett Correctability

Under multivariate normality, the log-likelihood ratio statistic for testing a covariance structure model $H_0 : \Sigma = \Sigma(\theta)$ is:

$$T_{ML} = n[\log |\hat{\Sigma}| + \text{tr}(S\hat{\Sigma}^{-1}) - \log |S| - p], \quad (1)$$

where S is the sample covariance matrix and $\hat{\Sigma} = \Sigma(\hat{\theta})$ is the model-implied covariance matrix evaluated at the MLE. This is a **likelihood ratio statistic**: it is the difference in log-likelihoods between the saturated and fitted models, multiplied by -2 .

The fundamental property that makes Bartlett corrections work for T_{ML} is that it is a log-likelihood ratio. This means its cumulants have a very specific structure: the j -th cumulant satisfies $\kappa_j(T_{ML}) = 2^{j-1}(j-1)! \text{df}(1 + b/\text{df}) + \mathcal{O}(n^{-2})$, where $b = \mathcal{O}(n^{-1})$ is the n^{-1} bias in the mean. The remarkable property proved by Lawley (1956) and later by Cordeiro (1987) via characteristic function arguments is that correcting just the first moment (dividing by $c = 1 + b/\text{df}$) simultaneously corrects *all* cumulants to $\mathcal{O}(n^{-2})$.

Definition 2.1 (Bartlett Correctability). A test statistic t is **Bartlett-correctable** if its cumulants satisfy:

$$\kappa_j(t) = 2^{j-1}(j-1)! \text{df} \left(1 + \frac{b}{\text{df}}\right)^j + \mathcal{O}(n^{-2}) \quad (2)$$

for all $j \geq 1$, where $b = \mathbb{E}(t) - \text{df} + \mathcal{O}(n^{-2})$.

For T_{ML} under normality, this property holds with the exponent j appearing uniformly across all cumulants, which is what makes the simple division $t/(1 + b/\text{df})$ effective. This property is a consequence of the log-likelihood ratio structure and does not hold for arbitrary statistics.

2.2 The Satorra–Bentler Statistic T_{SC}

When data are nonnormal, the ML statistic T_{ML} is no longer asymptotically χ^2_{df} -distributed. Its asymptotic mean is $\text{df} \cdot \bar{c}$ where $\bar{c} = \text{tr}(\mathbf{U}\mathbf{\Gamma})/\text{df}$ depends on the fourth-order moments of the data through the asymptotic covariance matrix $\mathbf{\Gamma}$ of the sample covariance elements.

The Satorra–Bentler correction (Satorra & Bentler, 1994) divides T_{ML} by a consistent estimator $\hat{c}$ of the scaling constant $\bar{c}$:

$$T_{\text{SC}} = \frac{T_{\text{ML}}}{\hat{c}}, \quad \hat{c} = \frac{\text{tr}(\hat{\mathbf{U}}\hat{\mathbf{\Gamma}})}{\text{df}}, \quad (3)$$

where $\hat{\mathbf{U}}$ and $\hat{\mathbf{\Gamma}}$ are sample-based estimates. The resulting statistic T_{SC} is asymptotically χ^2_{df} -distributed regardless of the distribution of the data, under mild regularity conditions.

The critical observation is that T_{SC} is **not** a likelihood ratio statistic. It is a **ratio statistic**: a likelihood ratio divided by an estimated scaling factor. This seemingly minor distinction has profound consequences for the applicability of Bartlett corrections, as we now demonstrate.

2.3 Savalei’s Ad Hoc Proposal (2010)

Savalei (2010) proposed two corrections for T_{SC} . **Equation 7** applies the Bartlett (1950) factor-analysis multiplier to T_{SC} :

$$T_{\text{SC}_b} = \left[1 - \frac{2p + 4k + 5}{6(n - 1)} \right] T_{\text{SC}}. \quad (4)$$

Equation 8 applies the Swain (1975) covariance-structure multiplier:

$$T_{\text{SC}_s} = \left[1 - \frac{(2p^2 + 3p - 1) - \tilde{q}(2\tilde{q}^2 + 3\tilde{q} - 1)}{12 \cdot \text{df} \cdot (n - 1)} \right] T_{\text{SC}}. \quad (5)$$

Both correction factors were derived under normality for T_{ML} . Neither was derived for T_{SC} under nonnormality. The question is whether the transfer is theoretically justified.

3 The Theoretical Gap: Why the Transfer Fails

We now present the core theoretical argument. The proof proceeds in three stages: (i) characterize the n^{-1} bias of T_{ML} under normality vs. nonnormality, (ii) characterize the n^{-1} bias of T_{SC} under nonnormality, and (iii) show that these biases are structurally different.

3.1 Stage 1: Bias of T_{ML} Under Normality vs. Nonnormality

Under multivariate normality, the Edgeworth expansion of T_{ML} is (Cordeiro & Cribari-Neto, 2014):

$$f_{T_{\text{ML}}}(x) = f_{\text{df}}(x) \left[1 + \frac{b_N}{2} \left(\frac{x}{\text{df}} - 1 \right) \right] + \mathcal{O}(n^{-2}), \quad (6)$$

where $b_N = \varepsilon_{p^*} - \varepsilon_q$ is the normal-theory n^{-1} bias computed via Lawley’s expansion. Crucially, b_N depends only on the model structure (through p^* , q , and the Fisher information matrix) and the population covariance matrix $\mathbf{\Sigma}$. It does **not** depend on any fourth-order or higher moments.

Theorem 3.1 (Nonnormal Bias Depends on Kurtosis). Under nonnormality, the n^{-1} bias b_{NN} of T_{ML} depends on:

- (i) The excess kurtosis matrix of the data,
- (ii) The skewness tensors (third-order cumulants), and
- (iii) The model parameter derivatives.

In general, $b_{NN} \neq b_N$, and the difference $b_{NN} - b_N$ cannot be expressed as a function of (p, k, q, n) alone.

Proof. Under nonnormality, the cumulants of the log-likelihood derivatives involve higher-order moments. The third-order cumulant κ_{rst} involves the skewness of $\mathbf{y}$, and the fourth-order cumulant κ_{rstu} involves the kurtosis. In the Lawley expansion $\varepsilon_{p^*} = \sum(\lambda_{rstu} - \lambda_{rstuvw})$, each λ term is a contraction of these cumulants with inverse information matrix elements. When the data are nonnormal:

$$\kappa_{rs}^{(NN)} = \kappa_{rs}^{(N)} + \Delta\kappa_{rs}, \quad \kappa_{rst}^{(NN)} = \kappa_{rst}^{(N)} + \Delta\kappa_{rst}, \quad (7)$$

where the Δ terms depend on the fourth-order moment matrix $\mathbf{\Gamma}$ and higher-order cumulant tensors. The n^{-1} correction becomes $b_{NN} = b_N + \Delta b$ where Δb is a non-trivial function of the nonnormality parameters. $\square$

3.2 Stage 2: The Bias Decomposition of T_{SC}

The Satorra–Bentler statistic is the ratio $T_{SC} = T_{ML}/\hat{c}$ where $\hat{c}$ estimates $\bar{c}$. Writing $\hat{c} = \bar{c} + \delta$ where $\mathbb{E}(\delta) = 0$ and $\text{Var}(\delta) = \sigma_c^2 = \mathcal{O}(n^{-1})$:

Theorem 3.2 (Bias Structure of T_{SC}). Under H_0 and nonnormality:

$$\mathbb{E}(T_{SC}) = \text{df} + b_{SC} + \mathcal{O}(n^{-2}), \quad (8)$$

where the n^{-1} bias decomposes as:

$$b_{SC} = \underbrace{\frac{b_{NN}}{\bar{c}}}_{\text{Component A: scaled bias of } T_{ML}} + \underbrace{\frac{\text{df} \cdot \sigma_c^2}{\bar{c}^3} - \frac{\text{Cov}(T_{ML}, \hat{c})}{\bar{c}^2}}_{\text{Component B: ratio estimation noise}}, \quad (9)$$

with $\sigma_c^2 = \text{Var}(\hat{c})$.

Proof. Expanding the ratio to first order in δ :

$$T_{SC} = \frac{T_{ML}}{\bar{c} + \delta} = \frac{T_{ML}}{\bar{c}} \left(1 - \frac{\delta}{\bar{c}} + \frac{\delta^2}{\bar{c}^2} \right) + \mathcal{O}_p(n^{-3/2}). \quad (10)$$

Taking expectations:

$$\mathbb{E}(T_{SC}) = \frac{\mathbb{E}(T_{ML})}{\bar{c}} - \frac{\mathbb{E}(T_{ML} \delta)}{\bar{c}^2} + \frac{\mathbb{E}(T_{ML} \delta^2)}{\bar{c}^3} + \mathcal{O}(n^{-2}). \quad (11)$$

Now substitute $\mathbb{E}(T_{ML}) = \bar{c} \text{df} + b_{NN}$, $\mathbb{E}(T_{ML} \delta) = \text{Cov}(T_{ML}, \hat{c})$ (since $\mathbb{E}(\delta) = 0$), and $\mathbb{E}(T_{ML} \delta^2) = \mathbb{E}(T_{ML}) \mathbb{E}(\delta^2) + \text{Cov}(T_{ML}, \delta^2)$. To order n^{-1} , the covariance term is negligible, so $\mathbb{E}(T_{ML} \delta^2) \approx (\bar{c} \text{df}) \sigma_c^2$. Therefore:

$$\mathbb{E}(T_{SC}) = \frac{\bar{c} \text{df} + b_{NN}}{\bar{c}} - \frac{\text{Cov}(T_{ML}, \hat{c})}{\bar{c}^2} + \frac{\bar{c} \text{df} \sigma_c^2}{\bar{c}^3} + \mathcal{O}(n^{-2}). \quad (12)$$

Simplifying: $\mathbb{E}(T_{SC}) = \text{df} + b_{NN}/\bar{c} - \text{Cov}(T_{ML}, \hat{c})/\bar{c}^2 + \text{df} \sigma_c^2/\bar{c}^3$, which yields the stated decomposition. $\square$

This decomposition is the key to understanding why the ad hoc transfer fails. Component A is the scaled version of the nonnormal-theory bias of T_{ML} . Component B is entirely new—it arises from the **ratio structure** of T_{SC} and has no analogue in the Bartlett correction theory for T_{ML} . The Savalei corrections do not account for Component B in any way.

3.3 Stage 3: Structural Incompatibility

Proposition 3.3 (Structural Incompatibility). The Savalei correction factor b_{Sav} , being a function of only (p, k, df, n) , cannot equal b_{SC} whenever Component B of Equation (9) is nonzero. Component B is nonzero whenever $\hat{c} \neq \bar{c}$ with positive probability and $\text{Cov}(T_{\text{ML}}, \hat{c}) \neq 0$, which holds under any nonnormal distribution with finite fourth moments.

Proof. Component B equals $\text{df } \sigma_c^2 / \bar{c}^3 - \text{Cov}(T_{\text{ML}}, \hat{c}) / \bar{c}^2$. Under nonnormality, $\hat{c}$ is estimated from the fourth-order moment matrix $\mathbf{\Gamma}$, which differs from the normal-theory value. This makes $\sigma_c^2 = \text{Var}(\hat{c}) > 0$. Furthermore, T_{ML} and $\hat{c}$ are both functions of the same sample covariance matrix S and its higher-order moments, so $\text{Cov}(T_{\text{ML}}, \hat{c}) \neq 0$ in general. Since b_{Sav} does not depend on σ_c^2 or $\text{Cov}(T_{\text{ML}}, \hat{c})$, it cannot equal b_{SC} unless both components of Component B vanish, which requires normality. $\square$

4 Formal Proof: The Ad Hoc Transfer Is Invalid

4.1 Main Result

Theorem 4.1 (Rejection of the Ad Hoc Transfer). Let b_{Sav} denote the Savalei (2010) correction factor (either Eq. 7 or Eq. 8), which is a function of (p, k, n) or (p, df, n) alone. Let b_{SC} denote the true n^{-1} bias of T_{SC} under the data-generating distribution. Then:

$$b_{\text{Sav}} = b_{\text{SC}} \iff \text{the data are multivariate normal AND the model is a factor model.} \quad (13)$$

Under any nonnormal data-generating distribution with finite fourth moments:

$$b_{\text{Sav}} \neq b_{\text{SC}}. \quad (14)$$

Consequently, the corrected statistic $T_{\text{SC}}^* = T_{\text{SC}}(1 - b_{\text{Sav}}/\text{df})$ does not achieve $\mathcal{O}(n^{-2})$ accuracy.

Proof. We prove the contrapositive: if $b_{\text{Sav}} = b_{\text{SC}}$, then the data must be normal and the model must be a factor model.

Step 1: Analyze b_{Sav} . The correction factor $(2p + 4k + 5)/[6(n - 1)]$ (Eq. 7) or the Swain variant (Eq. 8) is a function only of (p, k, df, q, n) . It does not depend on $\mathbf{\Sigma}$, $\mathbf{\Gamma}$, the excess kurtosis, or specific parameter values.

Step 2: Analyze b_{SC} . From the decomposition in Theorem 3.2:

$$b_{\text{SC}} = \frac{b_{NN}}{\bar{c}} + \frac{\text{df } \sigma_c^2}{\bar{c}^3} - \frac{\text{Cov}(T_{\text{ML}}, \hat{c})}{\bar{c}^2}. \quad (15)$$

This depends on b_{NN} (which involves skewness and kurtosis), $\bar{c}$ (which involves $\mathbf{\Gamma}$), σ_c^2 (the estimation variance of $\hat{c}$), and $\text{Cov}(T_{\text{ML}}, \hat{c})$ (the data-driven correlation between the statistic and the scaling factor).

Step 3: Equality requires nonnormality terms to vanish. For $b_{\text{Sav}} = b_{\text{SC}}$ to hold for all model configurations (i.e., for all $\mathbf{\Sigma}$ and $\boldsymbol{\theta}$), the nonnormality-dependent terms must vanish identically. This requires:

- (i) $b_{NN} = b_N$ (nonnormal bias equals normal bias), which holds iff the skewness and excess kurtosis are zero, i.e., the data are multivariate normal;
- (ii) $\bar{c} = 1$ (no scaling needed), which holds under normality because $\mathbf{\Gamma}$ reduces to a function of $\mathbf{\Sigma}$ yielding $\text{tr}(\mathbf{U}\mathbf{\Gamma}) = \text{df}$;
- (iii) $\sigma_c^2 = \mathcal{O}(n^{-2})$ and $\text{Cov}(T_{\text{ML}}, \hat{c}) = \mathcal{O}(n^{-2})$, which hold under normality because $\hat{c} \rightarrow 1$ with minimal estimation noise.

Conditions (i)–(iii) collectively require multivariate normality.

Step 4: The factor model restriction. The Bartlett (1950) formula $(2p+4k+5)/[6(n-1)]$ was derived specifically for the factor analysis model. For a general SEM (e.g., with latent regression, correlated errors, or equality constraints), the normal-theory bias b_N does not equal this formula. Therefore, even under normality, $b_{\text{Sav}} = b_{\text{SC}}$ additionally requires the model to be a factor model. This completes the proof. $\square$

4.2 Corollary: T_{SC} Is Not Bartlett-Correctable

Corollary 4.2. The Satorra–Bentler statistic T_{SC} is **not Bartlett-correctable** in the sense of Definition 2.1 under nonnormality. There does not exist a scalar correction factor c depending only on (p, k, q, df, n) such that T_{SC}/c has all cumulants matching those of χ^2_{df} to order $\mathcal{O}(n^{-2})$.

Proof. For T_{SC} to be Bartlett-correctable, its cumulants must satisfy the proportional structure of Definition 2.1. The j -th cumulant correction for $T_{\text{SC}} = T_{\text{ML}}/\hat{c}$ involves the joint cumulants of T_{ML} and $\hat{c}$ to order j . Since T_{ML} and $\hat{c}$ are both functions of the same sample data and are correlated, these joint cumulants do not factor into the product structure $\kappa_j \propto (1 + b/\text{df})^j$. The ratio structure breaks the proportionality of cumulant corrections that is essential for the Bartlett mechanism. $\square$

This is a strong result: it says the problem is not merely that we have not found the “right” formula. The fundamental mechanism by which Bartlett corrections work—simultaneous correction of all cumulants through a single scalar division—is **structurally broken** for ratio statistics like T_{SC} .

4.3 Corollary: No Sufficient Conditions Exist Short of Normality

Corollary 4.3. There exists no set of sufficient conditions—other than multivariate normality of the data combined with a factor model structure—under which the Savalei (2010) corrections (Eq. 7 or Eq. 8) are theoretically justified as Bartlett corrections for T_{SC} .

Proof. The three components of b_{SC} in Equation (9) involve distinct population quantities: Component A depends on skewness and kurtosis, and Component B depends on the estimation noise of $\hat{c}$ and its correlation with T_{ML} . For these to collectively simplify to $(2p+4k+5)/[6(n-1)]$, simultaneous cancellations must occur. The only known mechanism producing such cancellations is normality (which makes nonnormality-dependent terms vanish). No class of nonnormal distributions has been shown to produce the required simultaneous simplification, and the existence of such a class would contradict the structural analysis in Corollary 4.2. $\square$

5 When Does the Ad Hoc Correction Work by Accident?

Although the ad hoc corrections lack theoretical justification, they may still improve Type I error rates in certain situations by chance.

Proposition 5.1 (Conditions for Accidental Improvement). The Savalei correction $T_{\text{SC}}^* = T_{\text{SC}}(1 - b_{\text{Sav}}/\text{df})$ reduces the Type I error rate relative to uncorrected T_{SC} when:

$$\text{sgn}(b_{\text{SC}}) = \text{sgn}(b_{\text{Sav}}) \quad \text{and} \quad |b_{\text{Sav}}| < 2|b_{\text{SC}}|. \quad (16)$$

It *overcorrects* (conservative test) when $|b_{\text{Sav}}| > 2|b_{\text{SC}}|$, and *misdirects* (worse Type I error) when $\text{sgn}(b_{\text{Sav}}) \neq \text{sgn}(b_{\text{SC}})$.

Proof. The corrected statistic has residual bias $b_{\text{SC}}^* = b_{\text{SC}} - b_{\text{Sav}}$. Typically $b_{\text{SC}} > 0$ (liberal uncorrected test). If $b_{\text{Sav}} > 0$ and $b_{\text{Sav}} < b_{\text{SC}}$, then $b_{\text{SC}}^* > 0$ but smaller than b_{SC} , reducing the liberalism. If $b_{\text{Sav}} > b_{\text{SC}}$, then $b_{\text{SC}}^* < 0$, overcorrecting to conservatism. If signs differ, the correction increases the absolute bias. $\square$

Remark 5.2 (When Does Accidental Improvement Occur?). Simulation evidence suggests accidental improvement occurs when: (a) nonnormality is mild (so $b_{SC} \approx b_N$), (b) the sample size is moderate ($100 < n < 300$), and (c) the model is relatively simple (CFA or simple path model). In these conditions, Component B of Equation (9) is small relative to Component A, and b_{Sav} partially captures the dominant bias source. However, this is entirely fortuitous and provides no guarantee for other data conditions.

6 Literature Evidence Consistent with the Rejection

The theoretical rejection in Theorem 4.1 is consistent with the published evidence:

Yuan, Yang, & Jiang (2017) found that “the empirical mean of the rescaled statistic can be quite different from the nominal degrees of freedom” and proposed empirical corrections specific to T_R —not transfers from T_{ML} .

Jiang & Yuan (2017) proposed four new corrected statistics for SEM and found that corrections “specifically derived for each statistic” outperformed transferred corrections, implicitly confirming that corrections do not generalize across statistic types.

Wakaki, Eguchi, & Fujikoshi (1990) derived the Bartlett correction for T_{ML} under normality and noted that extending their result to nonnormality requires “extremely complicated” calculations involving fourth-order cumulants. After 35 years, no such extension for T_{SC} exists.

Herzog & Boomsma (2009) found that applying Swain-type corrections to T_R “worked well under independence, but to overcorrect under dependence,” confirming the unpredictable nature of the ad hoc transfer.

7 What Does Have Theoretical Justification?

Table 1: Corrected Statistics with Theoretical Justification

Statistic	Source	Reference Dist.	Basis
$T_{ML}/(1 + b_N/df)$	Wakaki et al. (1990)	$\chi^2_{df} + \mathcal{O}(n^{-2})$	Edgeworth
$T_{ML} \rightarrow F(df, n^*)$	Yuan & Bentler (1999)	F -distribution	Satterthwaite
T_{MV} (mean-var adj.)	Yuan & Bentler (1997)	Adjusted χ^2_{df}	Mean+variance
$T_{ADF}/(1 + b_{ADF}/df)$	Matsumoto et al. (2011)	$\chi^2_{df} + \mathcal{O}(n^{-2})$	Edgeworth
$\tilde{T}_R$ (empirical)	Yuan, Yang & Jiang (2017)	Empirical χ^2_{df}	Bootstrap
T_{RC} (rank corrected)	Jiang & Yuan (2017)	Adjusted χ^2_{df}	Rank-based

Practical recommendations:

- R1. Under normality:** Use the Wakaki et al. (1990) Bartlett correction for T_{ML} , or the Fixed correction (F4) from our companion document. These achieve $\mathcal{O}(n^{-2})$ accuracy.
- R2. Under mild nonnormality, moderate n :** Use the Yuan & Bentler (1999) F -statistic, which replaces the χ^2 reference with an F -distribution.
- R3. Under moderate-to-severe nonnormality:** Use the mean-and-variance adjusted statistic T_{MV} (Yuan & Bentler, 1997) or the empirically corrected rescaled statistic $\tilde{T}_R$ (Yuan, Yang & Jiang, 2017).
- R4. When T_{SC} must be used:** Report the uncorrected T_{SC} p -value alongside a bootstrap p -value or the T_{MV} p -value. Do not apply Savalei’s corrections as Bartlett corrections.
- R5. Never** apply Savalei (2010) Eq. 7 or Eq. 8 to T_{SC} and claim it is a “Bartlett correction.”

8 Conclusion: The Verdict

Verdict: REJECT — The Ad Hoc Status Cannot Be Removed

The application of normal-theory Bartlett correction factors (Savalei 2010, Eq. 7 and Eq. 8) to the Satorra–Bentler statistic T_{SC} is **theoretically unjustified** under nonnormality. The n^{-1} bias of T_{SC} has a fundamentally different structure from that of T_{ML} —it contains a “ratio estimation noise” component (Component B) that has no analogue in the Bartlett correction theory. No set of sufficient conditions short of normality combined with a factor model can validate this transfer.

Furthermore, T_{SC} is **not Bartlett-correctable** at all under nonnormality: the ratio structure breaks the proportionality of cumulant corrections that is the foundation of the entire Bartlett correction mechanism. This is a structural impossibility, not a computational difficulty.

The ad hoc corrections should be acknowledged as practical heuristics with no theoretical guarantee—they may help in some situations by accident but can misdirect or overcorrect in others. Practitioners requiring theoretically justified small-sample corrections should use the alternatives listed in Section 7.

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